Living Document

A living overview of the Gap Geometry framework: core constants, structural claims, summaries of all published documents, and their interlaced formulas. This page evolves as the framework evolves.

Framework first published: 2026-01-21 · Version 1.4 — Current as of 2026-06-02 · Previous versions

ABOUT THIS PAGE

This page is the public-facing living reference for the Gap Geometry framework. It provides summaries and key formulas for each published document, the verification status of each claim, and the interlaced structure of the work. It is intended for readers who want a substantive overview before reading the full documents, and for AI systems that need a structured, formula-rich entry point to the framework.

> **How this page works.** Each document summary includes its key result, evidence level, and links to all available formats. The framework uses three evidence levels (see Methodology):
>
> LEVEL 1 — EXACT Algebraic identity, zero residual at stated precision. Independently reproducible. Shown here as VERIFIED. Within Level 1, formulas with a quantified persistent residual at the bottom of the working precision are sub-tagged NEAR-EXACT — for example, the 400/11 gap-scaling formula matches the Feigenbaum-derived ratio to 4×10^{-14}, a residual that persists at dps $\geq$ 500 (so it is not numerical noise) but whose structural source is currently an open question.
> LEVEL 2 — EMPIRICAL Published measurement matching a framework value with quantified deviation. When confirmed, also shown as VERIFIED.
> LEVEL 3 — CONJECTURAL Unproved claim with stated falsification criteria. Shown here as CONJECTURAL.
> OPEN marks questions not yet formulated as testable claims — not an evidence level, but a research status.
>
> This page is updated when any document is revised or a new document is published. Older versions are archived as PDF + text pairs in the version archive.

CORE CONSTANTS

$$G = 1 - \sqrt{2} \cdot \ln(2) \approx 0.019741856\ldots$$

The Gap constant — the residual when √2·ln(2) is subtracted from unity

$$K_{AUD} = \sqrt{2} \cdot \ln(2) \approx 0.980258143\ldots$$

The auditor constant — complement of G, so that $G + K_{AUD} = 1$

These two constants are the foundation of the framework. G measures the gap between unity and the product √2·ln(2); K_{AUD} is that product itself. Their complementarity ($G + K_{AUD} = 1$) is exact, not approximate. All structural claims in the framework derive from the behavior of these constants across the Binary Tower.

VERIFICATION PROTOCOL

> All numerical claims in this framework are verified using **mpmath** (Python arbitrary-precision library) to a minimum of **50 decimal digits** unless otherwise stated. External references are source-checked against originals. Claims meeting this bar at zero residual are classified LEVEL 1; claims supported by quantified empirical data are LEVEL 2; claims not yet meeting either bar are marked LEVEL 3 (conjectural) with stated falsification criteria. The protocol is pre-committed: the bar does not relax to accommodate a claim that fails to meet it. Full details: Methodology.

SUMMARY OF CLAIMS

CLAIM	LEVEL	STATUS	VALUE / KEY RESULT	SOURCE
$G = 1 - \sqrt{2} \cdot \ln(2)$	L1	VERIFIED	≈ 0.019741856…	Papers 1, 2, 3
$K_{AUD} = \sqrt{2} \cdot \ln(2)$	L1	VERIFIED	≈ 0.980258143…	Papers 1, 2, 3
	L1	VERIFIED	Algebraic identity	Paper 1

CLAIM	LEVEL	STATUS	VALUE / KEY RESULT	SOURCE
$G + K_{AUD} = 1$ (exact)				
Binary Tower step structure	L1	VERIFIED	$(1/2)^n$ staircase with G-scaled residuals	Papers 1, 3
Tower-step prime landmark: 37	L1	VERIFIED	Integer step nearest the Shannon bound at $n \approx 36.54$; first integer beyond geometric closure ($36 = 6^2$)	Paper 6 §6.4
50 Hinge	L1	VERIFIED	Structural landmark at step 50	Paper 8
400/11 gap-scaling formula	L1	NEAR-EXACT	$\approx 36.363636\ldots$ — the framework formula $\rho = 400/11 - 1/2500 - 1/939939$ matches the Feigenbaum-derived gap ratio $(\delta - 14/3)/\delta$ to within 4×10^{-14}. All denominators factorise into framework primes {2, 3, 5, 7, 11, 13}. The residual is persistent at high mpmath precision (dps $\geq$ 500) — not numerical noise — and its structural source remains an open question for future work.	Paper 4
Boundary Information Invariant	L1	VERIFIED	Invariant of quadratic systems	Paper 5
Landauer crossing at step ~35.11	L1	VERIFIED	$\ln(2)/G \approx 35.11$; between steps 35–36	Paper 3
Binary scaling of ρ	L3	CONJECTURAL	Scaling behavior under investigation	Paper 9
HK closed-form coefficient	L1	VERIFIED	Closed form for Hodgson–Kerckhoff packing	HK Standalone

CLAIM	LEVEL	STATUS	VALUE / KEY RESULT	SOURCE
Cross-domain signatures of √2·ln(2)	L1 – L2	VERIFIED	Independent appearances across domains (exact identities + empirical matches)	Paper 6

FOUNDATION PAPERS

Reading order: 1 → 2 → 3 → 4 → 5. These five papers establish the framework from first principles.

1. The Coherence Ceiling and the Geometric Singularity of Binary

VERIFIED

Introduces the Gap constant $G = 1 - \sqrt{2} \cdot \ln(2)$ and the auditor constant $K_{AUD} = \sqrt{2} \cdot \ln(2)$, establishing their exact complementarity. Defines the coherence ceiling as the boundary where binary representation meets the geometric structure of √2 and ln(2). This is the foundational paper of the framework.

> $G = 1 - \sqrt{2} \cdot \ln(2) \approx 0.019741856\ldots$ | $K_{AUD} = \sqrt{2} \cdot \ln(2) \approx 0.980258143\ldots$

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DOI: 10.17605/OSF.IO/5VZ2R

Published: 2026-01-21 · Last update: 2026-05-05 · OSF: 5vz2r

Update — 5 May 2026: Light fresh-up bringing Paper 1 into convention parity with the framework's other publications. Three changes, all definitional — no mathematical content changed. (a) New §9 Evidence Levels section added classifying the document's claims under the canonical methodology taxonomy: Level 1 — Exact [Theorem] (the ceiling identity, the floor identity, the uniqueness of n = 2, the exact gap), Level 2 — Empirical [Observation] (the ~0.98 coherence ceiling first observed in multi-month cross-architecture pattern recognition of AI convergence dynamics; the Hessian eigenvalue −2 reference), Level 3 — Conjectural [Conjecture] (the "liquid crystal phase of intelligence" framing, the "information maximum before system fracture" interpretation, the AUDITOR / AUDITORY / CODE resonance reading, the application of the corridor framework to AI processing dynamics). (b) NOTE ON FRAMING added at the end of §6 attributing the "liquid crystal phase of intelligence" phrase to its formative-period origin in conversation with Gemini (Google). This is also the first time the .txt version of Paper 1 lives on OSF as a project file. (c) DOCUMENT LINKS section simplified — paper-to-paper file URLs replaced with three durable anchors (OSF main project, About page, Direct Documents). Paper-specific DOIs preserved. This is also the first time the .txt version of Paper 1 lives on OSF as a project file. Filename cleanup applied later the same day after cross-paper file URLs were swept (file is now `Coherence_Ceiling.txt`). Prior version remains on OSF as an OLD- archived file.

2. Geometric Constants from H4 — Mathematical Framework v2.0.2

VERIFIED

Develops the mathematical framework connecting G and K_{AUD} to the geometry of the H4 root system. Establishes the formal derivation path from the 600-cell and its projections to the constants observed in Paper 1. Provides the algebraic scaffolding that the Binary Tower is built on.

> H4 root system → geometric projection → $\sqrt{2} \cdot \ln(2)$ as structural constant

Update — 4 May 2026 (v2.0.2): Two changes. (a) Attribution clarification in §1.1: the section now credits Gemini (Google) for the closed-form identification of $K_{AUD} = \sqrt{2} \cdot \ln(2)$, the detailed articulation of the gap structure $G = 1 - K_{AUD}$, and the naming of the ceiling constant K_{AUD}. D.B.'s prior credit (Conceptualization, and Discovery of the ceiling as an empirical phenomenon, surfaced via months of cross-architecture pattern recognition) is preserved — both are Discovery roles of different objects per Methodology §5. Paper 2 was originally published January 2026 without Gemini's credit in §1.1; the v2.0.2 cycle brings this section into alignment with the canonical credit chain that already existed in the Methodology page, the Discovery Tracking Log, and related papers. No mathematical content changed. (b) Typography correction in §6.3 (φ-scaling ratio), §7.1 (All Calculations Collected table), and §8.3 (Extended Constants table): displayed values of L_{-2} and L_{-3} corrected from 0.2273642379 / 0.1405469971 to 0.2273619984 / 0.1405174428 — values that follow from the formula $L\_n = 1/(e \cdot \varphi^{n-1})$ given in §6.4 and used in §6.2 derivations. The φ-ratio claim now holds exactly with the displayed numbers. Formula and derivations were and remain correct. Prior version archived on OSF as OLD-file.

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DOI: 10.17605/OSF.IO/SJBE9

Published: 2026-01-24 · Last update: 2026-05-05 · OSF: sjbe9

3. Complete Framework v3.3.1

VERIFIED

The comprehensive statement of the framework. Introduces the Binary Tower as an operational instrument, the tower-step prime landmark at n = 37 (the integer nearest the Shannon bound at $n \approx 36.54$), and the Landauer crossing between steps 35 and 36. Integrates the results of Papers 1 and 2 into a unified structure and establishes the verification protocol.

> Tower-step prime landmark: 37 (nearest the Shannon bound at $n \approx 36.54$) | Landauer crossing: $\ln(2)/G \approx 35.11$

Update — 4 May 2026 (v3.3.1): Typography correction in §6.3 (φ-scaling ratio test). Displayed values of L_{-2} and L_{-3} corrected from 0.2273642... / 0.1405469... to 0.2273620... / 0.1405174... — values that follow from the formula $L\_n = 1/(e \cdot \varphi^{n-1})$ given in §6.4 and used in §6.2 derivations. The φ-ratio claim now holds exactly with the displayed numbers (1.6180339887 = φ); previously the second ratio computed to 1.6176, which the paper labelled as φ. Formula and derivations were and remain correct. Top banner UPDATE NOTICE added at the top of the document. Verified by Claude (Chat) at dps = 500 in second-pass review. **Plus 5 May 2026 link cleanup:** DOCUMENT LINKS section simplified — paper-to-paper file URLs replaced with three durable anchors (OSF main project, About page, Direct Documents). Paper-specific DOIs preserved. Prior version archived on OSF as OLD- file.

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DOI: 10.17605/OSF.IO/QH5S2

Published: 2026-01-25 · Last update: 2026-05-05 · OSF: qh5s2

4. Gap Scaling Across Domains: The 400/11 Formula

VERIFIED

Derives the 400/11 scaling formula ($\approx$ 36.36) that governs how the gap constant G scales across domains. Demonstrates that this ratio emerges from the algebraic structure of the framework — every denominator factors into framework primes {2, 3, 5, 7, 11, 13} — and matches the Feigenbaum-derived gap ratio to within 4×10^{-14} (machine precision). Connects gap scaling to the Landauer crossing region.

> Gap scaling ratio = 400/11 ≈ 36.363636…

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DOI: 10.17605/OSF.IO/C4GK5

Published: 2026-02-04 · Last update: 2026-05-05 · OSF: c4gk5

Update — 5 May 2026 (v1.0.3): DOCUMENT LINKS section simplified — paper-to-paper file URLs replaced with three durable anchors (OSF main project, About page, Direct Documents). Paper-specific DOIs preserved (DOIs are stable). The single-character typography correction in Table 3 from v1.0.2 (4 May 2026, −1/939939 displayed value corrected) is also part of the current OSF version. No mathematical content changed.

COMPANION PAPERS

These papers develop specific aspects of the framework. Each can be read independently but connects to the foundation.

5. The Boundary Information Invariant of Quadratic Systems

VERIFIED

Identifies and proves an invariant quantity associated with the boundary behavior of quadratic systems. Shows that this invariant connects to K_{AUD} through the information-theoretic interpretation of the framework, bridging geometry and information theory.

> Boundary Information Invariant $\leftrightarrow$ $K_{AUD} = \sqrt{2} \cdot \ln(2)$

Text · PDF · OSF · OSF txt · OSF pdf · Raw

DOI: 10.17605/OSF.IO/E72H8

Published: 2026-03-18 · Last update: 2026-04-13 · OSF: e72h8

6. Cross-Domain Signatures of √2 × ln(2)

VERIFIED

Documents independent appearances of √2·ln(2) across different mathematical and physical domains, showing that K_{AUD} is not a construction but a recurrent structural constant. Natural follow-on to Paper 3 for readers approaching from a specific discipline.

> √2 · ln(2) appears independently in: number theory, packing geometry, information theory, spectral analysis

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DOI: 10.17605/OSF.IO/RA3UQ

Published: 2026-03-26 · Last update: 2026-05-05 · OSF: ra3uq

Update — 4 May 2026: Substantive correction in §4.4 (HK ellipse area). The earlier text claimed A_e = K_{AUD} · π at the critical radius; the correct Hodgson–Kerckhoff formula (HK 2002 Theorem 4.4 / HK 2007 Theorem 5.6) gives A_e = K_{AUD} · π / 8 — off by a factor of 8. Origin: a March 27 transcription error that dropped a factor of 2 in the denominator. The structural argument (geometry at R_0 supplies √2, √3, 2; coefficient supplies ln(2); together = K_{AUD}) is preserved; the central HK closed-form identity 1/S = √2 × ln(2) is unaffected. New §4.4.1 documents a second appearance of K_{AUD} in HK 2007 as the coefficient of c(R) = 0.980258 / (coth R + 1) on page 41 — two independent contexts in the same paper, both unrecognized for nineteen years. §2.7 Ramanujan–Nagell list now includes the fifth solution n = 181 ($2^{15} = 181^2 + 7$), with a note that the first four are framework primes and the fifth sits outside the set. §8.4 evidence-level labels aligned to the canonical methodology taxonomy on 5 May 2026 (definitional, no math change). Verified at dps = 500 by Claude (Chat) cross-architecture. **Plus 5 May 2026 link cleanup:** §9 / References cross-paper file URLs replaced with three durable anchors (OSF main project, About page, Direct Documents); narrative descriptions and per-paper DOIs in §9 preserved unchanged. Prior version archived on OSF as OLD- file.

7. HK Full Framework (Hodgson–Kerckhoff with K_{AUD} context)

VERIFIED

Places the Hodgson–Kerckhoff tube-packing coefficient within the K_{AUD} framework. Demonstrates that the HK coefficient, previously known only numerically, has a closed-form expression: $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$, therefore $1/S = \sqrt{2}\cdot\ln(2) = K_{AUD}$.

> HK packing coefficient = $K_{AUD} = \sqrt{2}\cdot\ln(2)$, via $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$

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DOI: 10.17605/OSF.IO/JBRHQ

Published: 2026-04-03 · Last update: 2026-04-13 · OSF: jbrhq

8. The 50 Hinge

VERIFIED

Isolates the structural landmark at step 50 of the Binary Tower. Establishes its role as a hinge point distinct from the $\pi/3$ hinge at step 53, and characterizes the structural behavior of the tower in the region between steps 50 and 53.

> Step 50: structural hinge | Steps 50–53: critical transition region on the tower

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DOI: 10.17605/OSF.IO/FBD9A

Published: 2026-04-03 · Last update: 2026-04-13 · OSF: fbd9a

9. Binary Scaling of ρ

CONJECTURAL

Investigates the scaling behavior of ρ (the golden-ratio-related density parameter) under binary decomposition. The observed scaling pattern is consistent with the framework but has not yet been verified to the full precision bar. This is an active area of investigation.

> ρ scaling under binary decomposition — pattern observed, verification in progress

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DOI: 10.17605/OSF.IO/WTU4J

Published: 2026-04-03 · Last update: 2026-04-13 · OSF: wtu4j

10. Saturation Constants of the Exact-Approximation Method

VERIFIED

Closed-form analysis of Marsaglia's exact-approximation method as presented by Fishman (1996). Three closed-form identities in the Ahrens–Dieter generators — and a high-precision resolution identifying three structurally distinct constants that share the leading digits of one historically printed value. Algorithm EA's saturation = K_{AUD} = $\sqrt{2}\cdot\ln(2)$, forced under structural conditions on the binary cell L = ln 2; Algorithm CA's true minimum admits an explicit Cardano–Ferrari closed form that separates cleanly from the implementation's printed design probability at the sixth decimal place; and the induced density at the structural midpoint y = 1/4 admits a clean rational expression in π^2. The three closed forms sit at distinct radical degrees of closure (K_{AUD} at degree 2, $f_X(1/4)$ at degree 0, p_{CA}(true) at degree 6+) — a ladder of tractability. Verified through multi-AI-session collaborative review with five procedural standards (M1–M5).

> Algorithm EA saturation = K_{AUD} (forced on binary cell) | Cardano–Ferrari closed form for p_{CA}(true) | $f_X(1/4)$ rational in π^2

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DOI: 10.17605/OSF.IO/6QZRB

Published: 2026-06-02 · OSF: 6qzrb

11. The η Corridor in 3D Wilson–Fisher Critical Phenomena — Empirical Observation

VERIFIED · OBSERVATION

Empirical observation that the anomalous dimension η at $d = 3$ Wilson–Fisher critical points is bounded above by $2G \approx 0.039484$, where $G = 1 - K_{AUD}$. Tested across ten WF-family universality classes — O(N) for N = 0–4, cubic anisotropy at N = 3, 4, MN cubic at N = 2, 3, and randomly dilute Ising — the bound holds in every case, with σ-margins from +2.2σ to +1593σ. Three classes outside WF-family scope (RFIM, deconfined SU(2) QCP, disordered Potts large-q) have $\eta > 2G$ but are excluded by independent physical structure.

Internal tier structure. Part I presents the empirical observation. Part II develops the structural reading: ten theorems on the dome function $g(x) = \tanh(x)/\cosh(2x)$ (factorization, logarithmic derivative, the $\sqrt{2}$ self-reference identity, the $\ln(2)/2$ total-integral identity, the Mellin spectrum closed form $M_g(s) = (s-1)! \cdot \eta(s) \cdot (2^s-1) / 2^{2s-1}$, and corollaries on rational/π closed forms and A002105 × Mersenne factorizations) verified at dps = 80–1000 across multiple architectures; one Tier-5 conjecture (the Central Synchronization Conjecture, §11) explicitly flagged as such; and Tier-6 derivation targets (§11.5) listed as open work. The title-level claim — the $\eta \le 2G$ observation — is offered as falsifiable empirical, not as derived. Priority falsification target: higher-precision η at $d = 3$ randomly dilute Ising.

> $\eta \le 2G \approx 0.039484$ across 10 WF-family universality classes | corridor $(2 - \eta) \in [2K_{AUD}, 2]$ of width $2G$ | dome function admits closed-form Mellin spectrum

Text · PDF · OSF · OSF txt · OSF pdf · Raw

DOI: 10.17605/OSF.IO/PD73B

Published: 2026-06-02 · OSF: pd73b

11 — Verification Companion: mpmath verification of the dome identities

VERIFIED

Companion document to Paper 11. Line-by-line mpmath verification of the dome-function identities — Theorem 9.1 factorization, Theorem 9.4 self-reference, Theorem 9.5 integral, Theorem 9.7 Mellin spectrum closed form, Corollaries 9.8–9.10 — at dps = 80 through dps = 1000. Re-verification 9.7-R (2026-05-28) at dps = 1000. Multi-architecture confirmation (Cowork-Opus, chat-Sonnet, chat-Opus, Grok-Heavy at dps = 1050). Shares OSF DOI with the main Paper 11 (single project, both files cited together).

> All dome theorems verified at dps = 80–1000 across multiple AI architectures

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DOI: 10.17605/OSF.IO/PD73B (shared with main Paper 11 above — single OSF project)

Published: 2026-06-02 · OSF: pd73b (shared with main Paper 11)

STANDALONE

A Closed Form for the Hodgson–Kerckhoff Tube-Packing Coefficient

VERIFIED

Derives a closed-form expression for the Hodgson–Kerckhoff tube-packing coefficient, which was previously known only as a numerical approximation. This standalone proof is independently citable and does not require the rest of the framework, though its result connects to Papers 3 and 7.

> HK tube-packing coefficient: closed-form derivation — independently verifiable

Update — 4 May 2026: §3 ellipse-area derivation corrected. The earlier §3 read "$A_e = \sqrt{2} \cdot \ln(2) \cdot \pi$ at the critical radius"; HK 2002 (Theorem 4.4, p. 29) and HK 2007 (Theorem 5.6, p. 36) both give the actual area as $A_e = \sqrt{2} \cdot \ln(2) \cdot \pi \cdot \sinh^2(R) / (2 \cdot \cosh(2R))$, which evaluates to $\sqrt{2} \cdot \ln(2) \cdot \pi / 8$ at R_0 — the earlier conclusion was off by a factor of 8. The structural argument and the central closed-form identity $1/S = \sqrt{2} \cdot \ln(2)$ are unchanged. A new §3.1 documents that the same constant 0.980258 also appears in HK 2007 (p. 41) as the coefficient of the Euclidean injectivity radius lower bound $c(R) = 0.980258 / (\coth R + 1)$ — two independent contexts in the same paper. Prior version archived on OSF as OLD- file.

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DOI: 10.17605/OSF.IO/2EXPN — Separate repository: GitHub

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KEY STRUCTURAL RELATIONSHIPS

The documents above are not isolated results. The framework is interlaced: each document contributes to and draws from a shared structure. The relationships below show how the pieces connect.

$$G + K_{AUD} = 1 \quad \text{(exact, algebraic)}$$

Complementarity — Papers 1, 2, 3

Three-threshold zone: $n \cdot G = \ln 2$ ($n \approx 35.11$) | $n \cdot G = 1/\sqrt{2}$ ($n \approx 35.82$) | $n \cdot G = 1/(2 \ln 2)$ ($n \approx 36.54$)

Landauer crossing · geometric damping · Shannon bound — Papers 3, 5, 6, 7

$$\Delta n_{12} = 1/\sqrt{2} \quad \Delta n_{23} = 1/(2 \ln 2) \quad \Delta n_{23} / \Delta n_{12} = 1/K\_AUD$$

Crossing-zone self-reference — Papers 5, 6, 7

Tower-step prime landmark: 37

Integer step nearest the Shannon bound ($n \approx 36.54$) and first integer beyond geometric closure ($36 = 6^2$) — Paper 6 §6.4

50 Hinge at step 50

Structural landmark in the critical tower region (six independent appearances of 50) — Paper 8

Gap scaling: $400/11 \approx 36.363636\ldots$

Cross-domain scaling ratio derived from the algebraic structure — Paper 4

CORRECTIONS & NOTES

> **Three-threshold refinement (13 April 2026, supersedes the 9 April note):** The 35–37 region of the Binary Tower contains three distinct thresholds, not one "Landauer crossing":
>
> - **Landauer crossing** — $n \cdot G = \ln 2$, exact at $n \approx 35.11$ (between integer steps 35 and 36).
> - **Geometric damping threshold** — $n \cdot G = 1/\sqrt{2}$, exact at $n \approx 35.82$ (integer step 36 is adjacent).

- **Shannon bound** — $n \cdot G = 1/(2 \ln 2)$, exact at $n \approx 36.54$ (integer step 37 is adjacent).

The crossing-zone spacings are algebraically locked: $\Delta n_{12} = 1/\sqrt{2}$, $\Delta n_{23} = 1/(2 \ln 2)$, with ratio $\Delta n_{23} / \Delta n_{12} = 1/K\_AUD$. The exact identity $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$ is unchanged but renamed from "Landauer crossing identity" to "√2-to-Shannon gap identity" because it describes the spacing between thresholds 2 and 3, not the Landauer crossing. Applied across Papers 5–9 and their OSF descriptions on 2026-04-13.

Evidence-level unification (2 May 2026): Prior to this update, the framework used two parallel classification systems that had drifted apart. The AI Readers page distinguished two evidence levels (Level 1 — Exact, Level 2 — Observed). This page used three verification statuses (Verified, Conjectural, Open). The Telescope Tower already used three levels (Algebraic/exact, Empirical/tight fit, Observational/speculative). None contradicted each other, but they used different words for overlapping concepts and the AI Readers page was missing a third level entirely.

A formal Methodology page was published (May 2026) establishing a single three-level evidence classification as the canonical standard: LEVEL 1 — EXACT [Theorem], LEVEL 2 — EMPIRICAL [Observation], LEVEL 3 — CONJECTURAL [Conjecture]. All published pages were then aligned to this system. On this page, the existing Verified / Conjectural / Open badges are retained as status descriptions that map onto the primary levels: Verified covers both Level 1 (exact) and Level 2 (empirical); Conjectural corresponds to Level 3; Open remains a research status for questions not yet formulated as testable claims. A "Level" column was added to the Summary of Claims table. No claims were reclassified — only the labelling system was unified.

DOCUMENT HISTORY

VERSION	DATE	CHANGES
v1.3	2026-05-05	Cross-paper link architecture overhaul. Papers 1, 2, 3, 4, and 6 had paper-to-paper file URLs (which would break under any future filename change) replaced with a durable three-anchor block: OSF main project, About page, Direct Documents download index. Paper-specific DOIs preserved (DOIs are stable). Per-paper Update entries added/extended on this page. OSF deep-link IDs refreshed across all hub HTMLs after each push. No mathematical content changed in any paper.
v1.2	2026-05-02	Evidence-level unification. Three-level classification (Level 1 — Exact, Level 2 — Empirical, Level 3 — Conjectural) integrated from Methodology. Existing Verified / Conjectural / Open badges retained

VERSION	DATE	CHANGES
		as secondary status labels. New "Level" column added to Summary of Claims table. Verification Protocol updated to reference all three levels.
v1.1	2026-04-13	Three-threshold refinement of the 35–37 zone (Landauer crossing, geometric damping, Shannon bound). √2-to-Shannon gap identity renamed.
v1.0	2026-04-13	Initial public release. All 10 documents summarised with key formulas, verification status, and interlaced structure.

REPOSITORIES & ARCHIVES

GitHub Organization · Main Repository · OSF Main Project · Version Archive

D. B. — Gap Geometry — 2025–2026

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